

Hybrid Hand-Actuated Espresso Machine



By

Team 1: Crema Crew

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TABLE OF CONTENTS

NOMENCLATURE	2
1. PROBLEM STATEMENT.....	4
2. FUNDAMENTALS	5
Josua Albertsson - Heating Coil Fundamentals	
Westley Baird - Linkage Fundamentals	
Gabriel Camacho - Brew Chamber Fundamentals	
Joseph Dwyer - Electrical Fundamentals	
Alexis Phipps - O-Ring Sizing and Piston Linkage Buckling Fundamentals	
3. FINAL DESIGN	9
4. ANALYSIS	14
Josua Albertsson - Heating Coil Analysis	
Westley Baird - Linkage Analysis	
Gabriel Camacho - Brew Chamber Analysis	
Joseph Dwyer - Electrical Analysis	
Alexis Phipps - O-Ring Sizing and Piston Linkage Buckling Analysis	
5. SYSTEM SPECIFICATION AND TEST PLAN.....	22
6. SCHEDULE.....	23
References.....	25

NOMENCLATURE

Symbol	Definition	Units
A	Cross-sectional area	mm ² , m ²
A _s	Surface area	mm ² , m ²
cp	Specific heat capacity of water	J/kg·K
d	Diameter	mm, m
D _i	Inner diameter	mm
D _o	Outer diameter	mm
E	Modulus of elasticity	GPa
F	Force	N, lbf
FS	Safety factor	Dimensionless
h	Convection coefficient	W/m ² ·K
I	Current	Amps
I _{area}	Area moment of inertia	mm ⁴
J	Polar moment of inertia	mm ⁴
k	Thermal conductivity	W/m·K
K	Effective length factor for buckling	Dimensionless
L	Length	mm, m
L _e	Effective column length	mm
m	Mass flow rate	kg/s
P	Pressure	Pa, bar, psi
P _{cr}	Critical buckling load	N
Q̇	Heat transfer rate	W
Q	Thermal energy	J
r	Radius	mm, m
r _i	Inner radius	mm
R	Electrical resistance	Ω

Symbol	Definition	Units
Re	Reynolds number	N/A
t	Wall thickness	mm
Temp	Temperature	°C, °F
Torque	Torque	N·m
V	Voltage	V
Vol	Volume	mL
W _c	O-ring groove clearance width	mm
W _o	O-ring cross-sectional width	mm
x	Linear displacement	mm
ΔT	Temperature difference	°C, K
μ	Dynamic viscosity	Pa·s
ρ	Density	kg/m ³
σ	Normal stress	MPa
σ _{cr}	Critical buckling stress	MPa
σ _{hoop}	Hoop stress	MPa
σ _{longitudinal}	Longitudinal stress	MPa
τ	Shear stress	MPa
τ _{max}	Maximum torsional shear stress	MPa
θ	Angular displacement	Degrees, radians
π	Mathematical constant pi	Dimensionless

PROBLEM STATEMENT

The goal is to produce a double shot of espresso consistently and per espresso industry standards. A single shot of espresso is defined to be 7-9 grams of finely ground coffee beans that is used to produce a 30mL “shot” of espresso and a double shot is 14-18 grams producing a 60mL shot. The extraction process uses 9 bars of pressure, equivalent to approximately 130 psi of pressure that presses water through a tamped “puck” of espresso contained within a 58mm portafilter. A fine mesh shower screen is used to evenly distribute water over the top of the puck during the extraction process. The tamping process uses anywhere from 20-30 lbs of pressure to compress the finely ground coffee beans, which is a process that occurs externally from the espresso machine and will be controlled by a calibrated tamper purchased during the testing phase of the build process. The temperature control of the extraction process is also considered with optimal temperature ranges for a shot of espresso being between 105-205 °Fahrenheit, which is to be controlled in the water heating process. Due to the unique qualities of each type of coffee bean and the interfacing between the brewing machine, the individual user often has to fine-tune the size of the coffee grinds, the amount of dry coffee, and the exact temperature of the water being pulled through the puck.

Design Specifications

- Industry Standard extracts a double shot of espresso at 9 Bar
- 450 mL water in a tank dispenses water at a variable temperature of 190-205°F to make 10 single shots of Espresso (30mL espresso extracted each shot + 50% by volume waste fluid)
- Easy actuation of lever arm approximately 10lbs by user
- 2 ounce of espresso $\pm 10\%$ vol. in 25-30 seconds
- Fits within 1ft x 2ft x 2ft area

FUNDAMENTALS

Heating Coil Fundamentals

Josua Albertsson

To heat the water, we will use an inline heater unit. Which will be comprised of a copper tube with a resistance wire wrapped around it. This will be treated as a constant heat flux problem to find the length of wire needed and length of the heating coil section. The heating section will need to be insulated the exact type of insulation will be determined after initial testing. The rate at which water is pumped through the coil governs the heat transfer mode, laminar or turbulent, which has a substantial impact on the rate of heat transfer [1].

$$\dot{Q} = \dot{m}c\Delta T \quad [2] \quad p = i^2 R \quad [1]$$

Linkage Fundamentals

Westley Baird

The linkage system undergoes significant forces through operation a buckling and torsion stress analysis will be performed. Both Von mises and Tresca criterion are considered. The Euler Formula for long column buckling is used and presented below and is applied to the long hiem joint linkage [3].

$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 E}{\left(\frac{KL}{r}\right)^2}$$

The Torsion stress analysis is done on the shaft undergoing significant torsional stress due to it being the main driver of the rest of the linkage. It is treated as a static loading under “worse case scenario” Below is the general equation for torsional stress applied to a round rod [4].

$$\tau = \frac{Tr}{J}$$
$$J = \frac{\pi d^4}{32}$$
$$\tau_{max} = \frac{16T}{\pi d^3}$$

Brew Chamber Fundamentals

Gabriel Camacho

The brew chamber is the core of the espresso maker. This is where all systems meet and combine. The chamber is essentially a slow-leaking pressure vessel. The portafilter filled with compressed coffee grounds serves as the bottom of the “pressure vessel.” Hot water at atmospheric pressure is pumped in and then subsequently the piston moves down to create pressure. Pressure is being constantly generated from the action of the piston going down and the pressure is released through the coffee going out of the bottom. The chamber is thin enough to be categorized as a thin-walled pressure vessel, so the calculations can be done with hoop and longitudinal stress calculations [5].

$$\sigma_{Hoop} = \frac{Pr_i}{t}$$
$$\sigma_{Longitudinal} = \frac{Pr_i}{2t}$$

Electrical Fundamentals

Joseph Dwyer

The electrical section is what gives this device uniqueness when comparing to prior hobbyist designs that can be found on the market. It's what allows the water to be heated up, pumped into the brew chamber, and possibly more features. However, we are deeply constrained by factors such as cost and wattage from wall outlets. It is important that electronics are optimized for efficiency purposes and budgetary to get the most out of the system. Due to this there will be varying voltages with different available amperages which will require conversions using watts law [6].

$$P = V \times I$$

Where P is wattage, V is voltage, and I is amperage. Beyond that there is a need for a PID control equation for temperature management that has not yet been developed.

O-Ring Fundamentals

Alexis Phipps

The O-ring is a critical component in the brewing chamber, attached to the piston head that pressure is ultimately built inside the chamber to brew the coffee. O-ring sizing is based primarily off fundamentals that can be found in most O-ring sizing handbooks, but the one used is Parker O-Ring [7]. The primary factors to account for when sizing O-rings and the piston sleeve that it will interface with are squeeze and stretch. Squeeze is the amount the O-ring is squished inside the channel it nests in and the inner face of the brew chamber wall. The formula and graphic are seen below in figure 1. For a dynamic scenario where the O-ring experiences repetitive motion, like our case, the percent squeeze must be between 10-25%. The other factor to account for is stretch. O-rings come in mostly standard, whole number sizes and as such the O-ring will need to stretch to fit around the piston within its channel. There should be 1-5% stretch seen in the O-ring upon installation. The formula and graphic are seen below in figure 2.

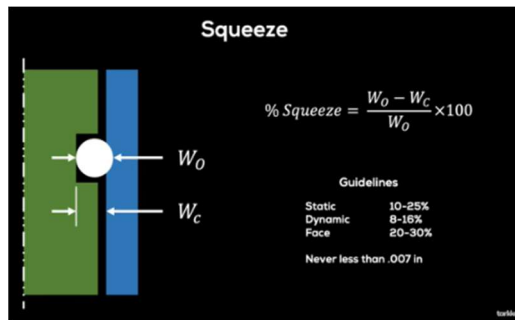


Fig. 1 Percent Squeeze formula [8]

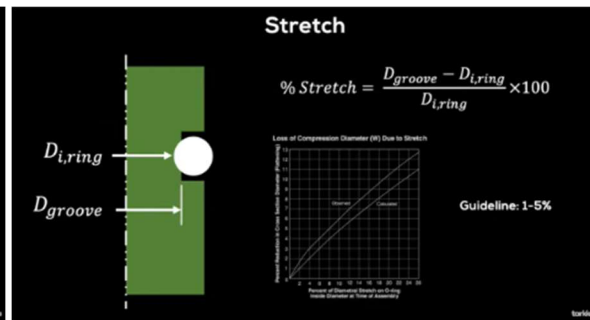


Fig. 2 Percent Stretch formula [8]

Buckling Fundamentals – Piston Linkage

Alexis Phipps

The piston linkage that connects the piston head to the frame linkage functions to translate the force into the pressure that ultimately presses the water through the finely ground beans to make the shot of espresso. In the linkage the only major forces incurred will be pure compression. Since the linkage is a pin-to-pin connection, there is an associated constant, k , that is equal to 1. To find the critical stress in the linkage the following formula is used, where E is the modulus of elasticity for the material, L is the unsupported length of the column, L_e is effective length based upon the pin type connection and the length, r is the smallest radius of gyration of the column equal to the square root of

the least moment of inertia over cross sectional area. The formulas outlining the calculations are below.

$$\sigma_{cr} = \frac{\pi^2 E}{\left(\frac{Le}{r}\right)^2} \quad Le = KL \quad r = \sqrt{I/A} \quad I = \frac{1}{12}bh^3 \quad \sigma = \frac{P}{A} \quad SF = \frac{\sigma_{cr}}{\sigma} \quad [9]$$

FINAL DESIGN

The final espresso machine design satisfies all stated design specifications while maintaining a compact, manufacturable, and serviceable assembly. The machine is designed to extract espresso at the industry standard pressure of 9 bar, produce a 2 oz espresso shot within 25 to 30 seconds, operate with approximately 10 lb of user input force, and fit within a 1 ft × 2 ft × 2 ft envelope.

The structural frame is constructed from 1/8 in thick 304 stainless steel sheet that is laser cut and precision bent into shape. The uprights supporting the brew chamber are tabbed and slotted into the base for accurate alignment and are stitch welded using TIG or laser welding processes to provide rigidity while minimizing weight. The integrated frame also houses the worm gear and linkage supports, allowing the power transmission system to blend seamlessly into the overall structure.

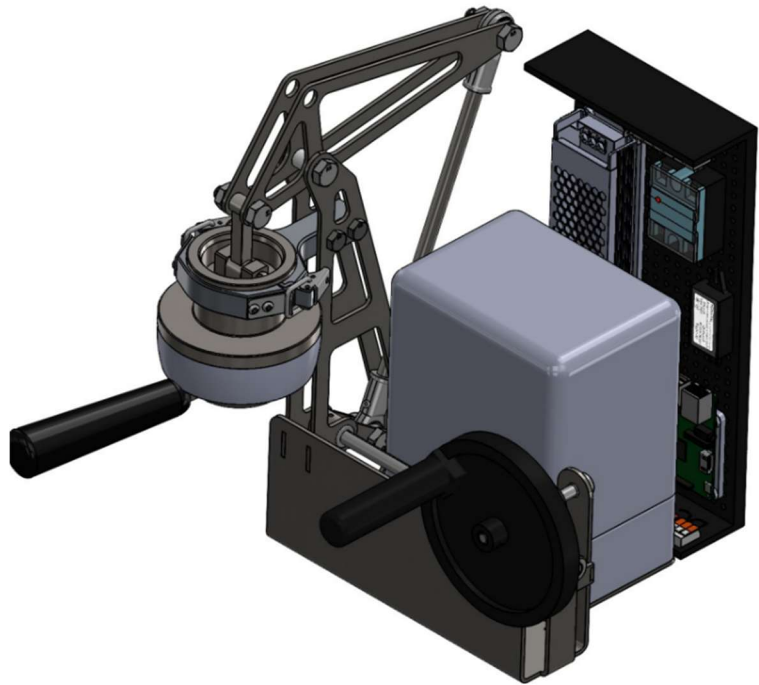
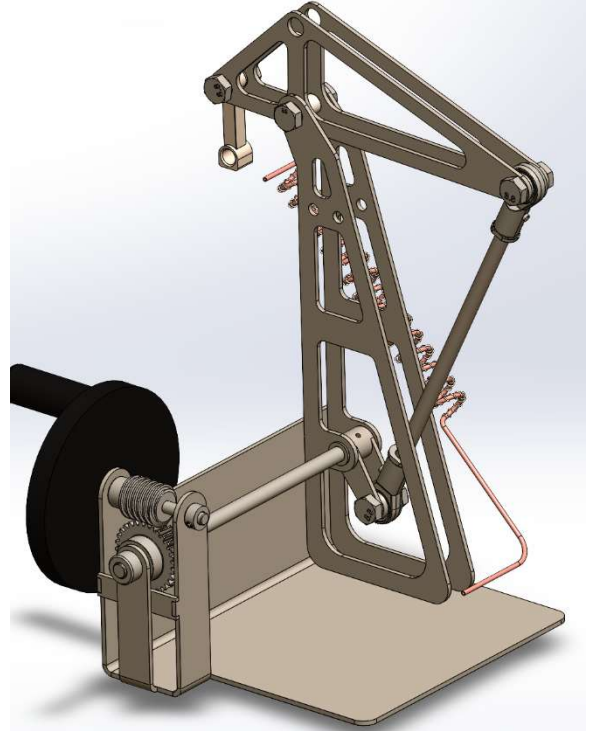


Figure 1: Isometric Full Assembly

To achieve the required brewing pressure of 9 bar, the machine uses a 30:1 worm gear reduction combined with 3:1 and 4:1 lever arm ratios. This mechanical advantage allows a user-applied force of approximately 10 lb at the handle to generate nearly 500 lb of force at the piston. The piston force pressurizes the brew chamber sufficiently to produce proper espresso extraction through the coffee puck contained in a standard 58 mm portafilter.

$$P = \frac{F}{A}$$

Figure 2: View showing full linkage system from handle to piston.



The brew chamber is also fabricated from stainless steel and includes integrated ports for heated water entry, a pressure gauge, and an air relief valve for safe operation and pressure monitoring. The chamber mounts to a commercially available 58 mm portafilter interface, enabling compatibility with off-the-shelf espresso accessories. The complete brew chamber assembly is secured to the uprights using an over-center clamp mechanism that allows rapid disassembly for cleaning and maintenance.

Figure 4: Over Center Clamp

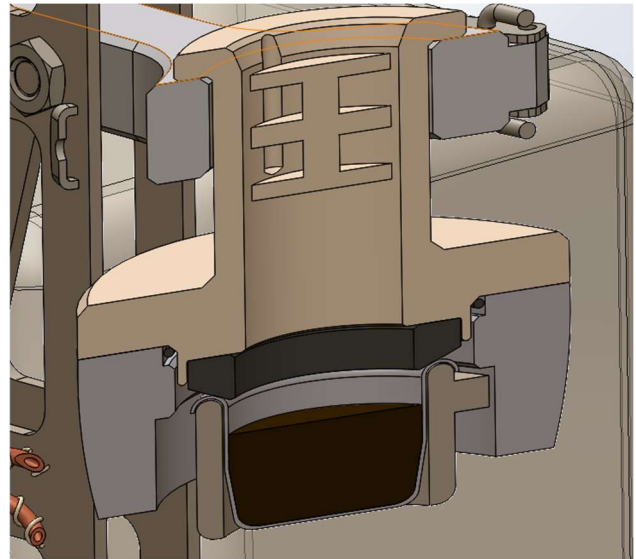
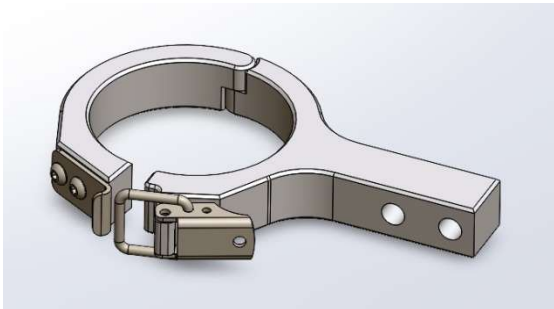


Figure 3: Brew Chamber Cross Section Showing Air Relive Valve.

The machine contains a 450 mL water reservoir, meeting the requirement to prepare approximately 10 single espresso shots when accounting for both extracted espresso volume and waste fluid generated during brewing. Water is pumped from the reservoir through a heated coil system activated by a push-button control. The heating coil raises the water temperature to approximately 96°C (205°F) before entering the brew chamber, satisfying the specified brewing temperature range of 190°F to 205°F.

$$T = 96^{\circ}\text{C} \approx 205^{\circ}\text{F}$$

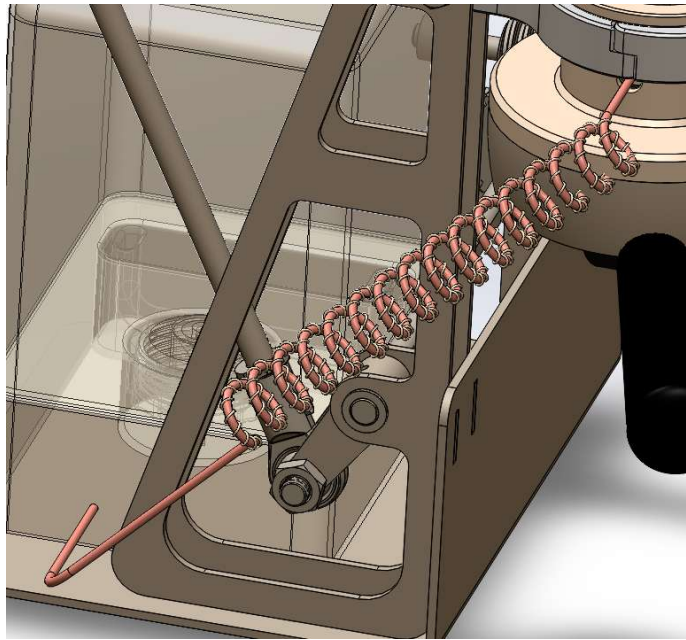


Figure 5: Water heating element coming from pump housing into brew chamber

The machine footprint is approximately 12 in wide, 9 in deep, and 15 in tall, which remains within the required dimensional constraint of 1 ft × 2 ft × 2 ft. The completed assembly has an estimated dry weight of approximately 14 lb, making the unit compact and portable while maintaining sufficient structural integrity for repeated high-pressure operation.

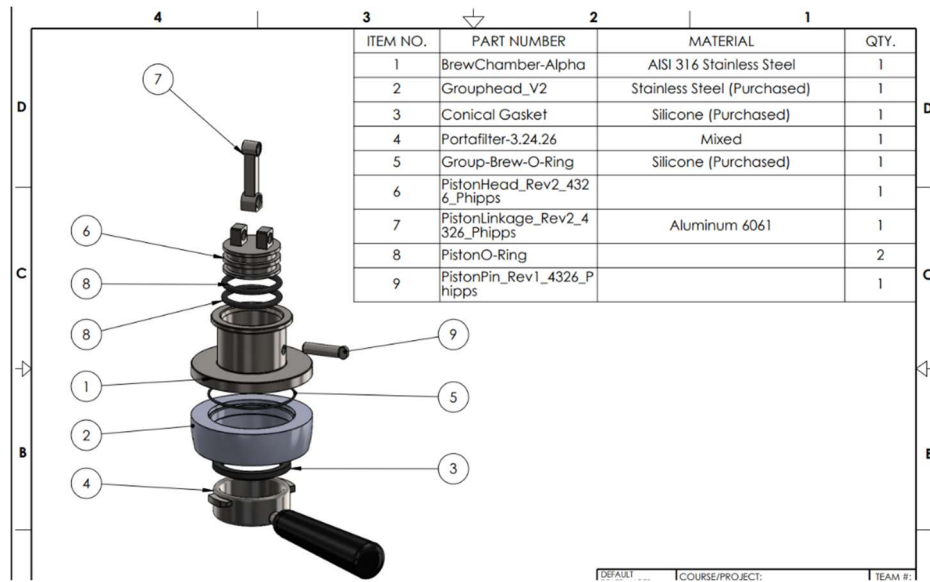
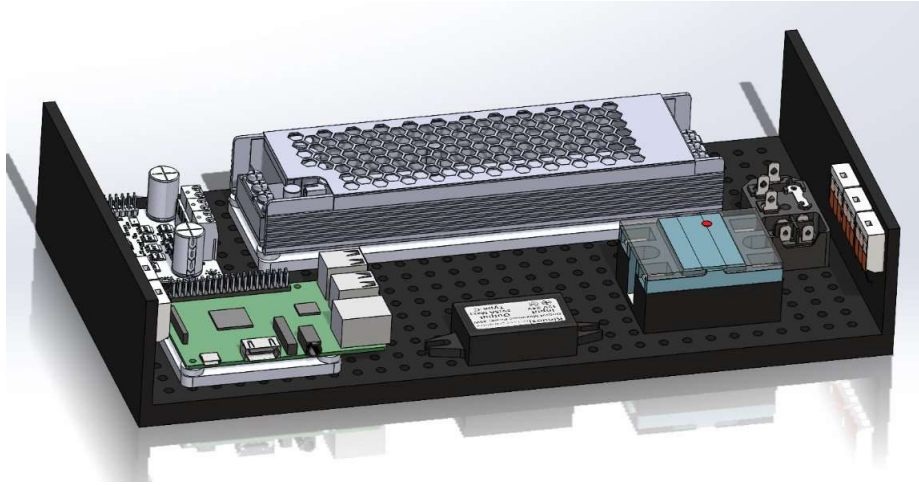


Figure 6: Brew Chamber Assembly

To incorporate the electrical demands of the system to heat the water and have an automated pumping system, several choices have been made in terms of engineering goals. To keep this device appliance compliant, it should be fused for 10A 120v to work with all American outlets. If needed, we can fallback onto high power outlet with 15A 120v fuses. Since our heating system requires rapid inline heating, our power output may demand straight AC heating. Additionally, to create precise repeatable fluid outputs, a peristaltic pump and Nema steppers will be implemented. This means the system requires a DC system for both the logic processor and stepper power which we opted to use 24V DC and 5V for logic. If there is additional time, we may also seek to add a motorized crank, automated

pressure curves and a touch screen to allow for smart espresso automation.



ANALYSIS

Josua Albertsson

5/21/26

Group 1: Crema Crew

Spring 2026

Heating Coil Analysis

Using the assumption that the copper coil's thermal resistance is negligible coupled with substantial insulation we assume that all of the energy generated will flow into the water. To determine if the fluid is going to heat evenly, we check the Reynolds number.

$$Re = \frac{4\dot{m}}{\pi D_i \mu} = \frac{4 * 0.0024}{\pi * 1.588E - 3 * 4.804E - 4} = 4007$$

Because the Reynolds number is above 4000 the water will be turbulent and we can assume there will be good heat transfer.

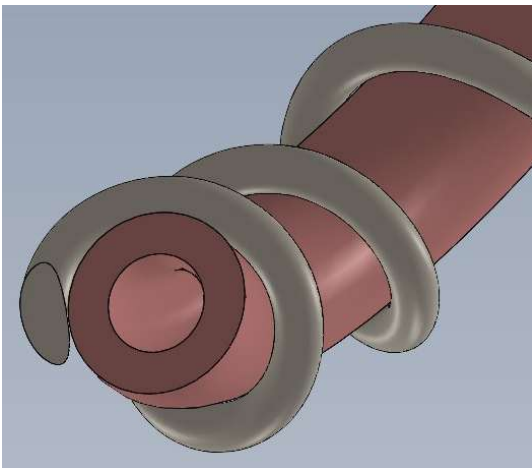
The energy required to heat water from 21 to 96 °C will be found.

$$Q = mc\Delta T = 0.60kg * 4.18 \frac{kJ}{kg * K} (96 - 21)K = 18.84 kJ$$

The time allowed for the water to heat is 25 seconds this yields the wattage.

$$\dot{Q} = \frac{j}{s} = \frac{18.84 kJ}{25s} = 754w$$

The resistance wire sourced has a resistance of 2.1ohms / meter using this we will find the needed length of wire (L_w) as well as the length of the needed pipe (L_p). Using the relation of length of wire to length of pipe.



$$\frac{L_w}{L_p} = \frac{\pi * 3.985}{0.81}$$

$$\dot{Q} = \frac{v^2}{R}$$

$$754w = \frac{120^2}{2.1 \frac{\Omega}{l_w} * \frac{\pi * 3.985 L_w}{0.81 L_p} * L_p}$$

$$L_p = 0.588m, L_w = 9.1m$$

Westley Baird
 5/20/26
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 Spring 2026
Linkage Analysis

The Linkage as the job of transferring and converting force from the hand crank to the piston, it does this through a system of multiple linkages and a worm and gear system, it is designed to provide approximately a 120 to 1 ratio allowing for 5lbs of force to produce nearly 600 before accounting for losses, due to this large 600 lb force transferring through the linkage system an analysis of the torsional shear force and a buckling analyses should be ran to ensure the linkage system is adequate.

Below is the buckling analysis done on the hiem joint rod all calculations were done in Excel to allow for auto population of results based on design changes.

Buckling Analysis

Westley Baird

$$F = P * A$$

$$F = 9\text{bar} * A$$

$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 * E}{\left(K * \frac{L}{r}\right)^2}$$

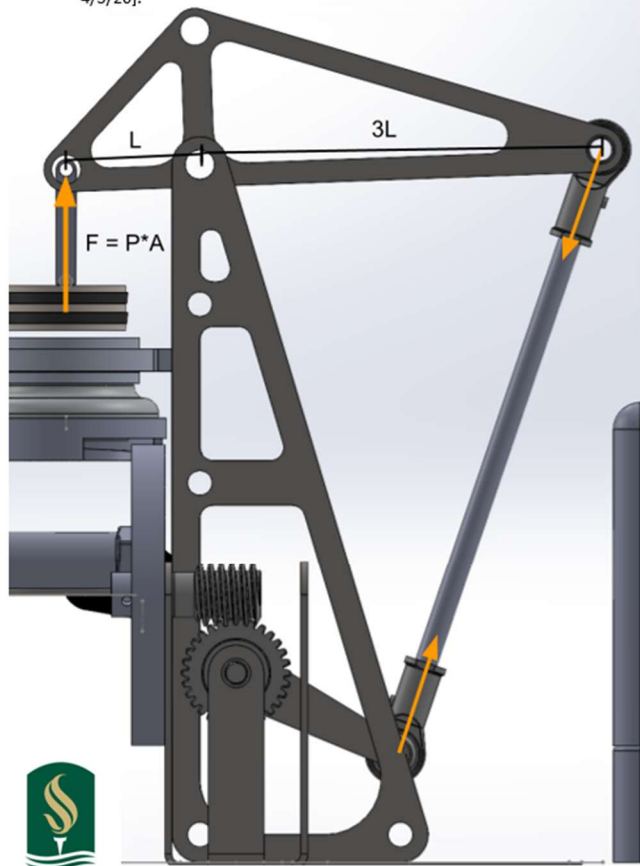
Material Properties 304 SS / Givens

Force (N)	890
Diameter (mm)	10
K Factor	1
Youngs Modulus (Gpa)	198
Rod Length (mm)	250

Calculations

$A = (\pi * d^2) / 4 \text{ (m}^2\text{)}$	0.0078538
Radius of Gyration = $d/4 \text{ (m)}$	0.0025
$I = \pi * d^4 / 64 \text{ (m}^4\text{)}$	4.909E-10
$P_{cr} = (\pi^2 * E * I) / (K * L)^2 \text{ (N)}$	15346.749
$FS = \text{Load} / P_{cr}$	17.244

[1] MechaniCalc, "Column Buckling", MechaniCalc, [Online].
 Available: <https://mechanicalcalc.com/reference/column-buckling>. [Accessed: 4/5/26].



Below is the Torsion Stress Analysis on the main driving rod for the linkage system, both Tresca and Von Mises criterion were considered. All calculations were done in excel to account for future iterations of the design. Here the Tresca SF is lower at 8.7 FS this is more than enough for this application

Torsion Stress Analysis

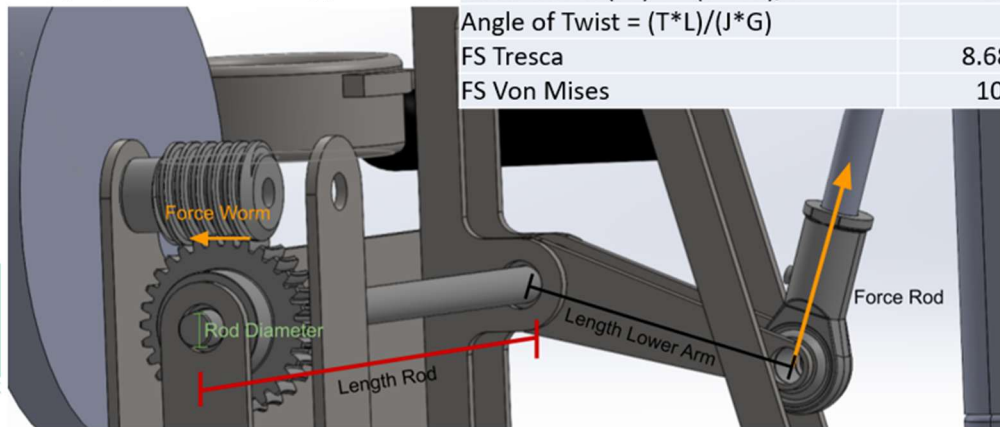
Westley Baird

$$\tau = \frac{Tc}{J}$$

$$J = \frac{\pi d^4}{32}$$

$$\phi = \frac{T * L}{J * G}$$

[1] Boston University, "Mechanics of Materials: Torsion", Boston University, [Online]. Available: <https://www.bu.edu/moss/mechanics-of-materials-torsion/>. [Accessed: 4/5/26].



Material Properties 1045 Steel

Yield Strength (S_y) Mpa	630
Shear Modulus (G) Gpa	80
Max Shear Stress τ_y (Tresca)	315
Max Shear Stress τ_y (Von Mises)	363.7306696

Givens

Linkage arm Length mm	40
Load N	890
Rod Length	200
Rod Diameter	0.1

Calculations

Torq applied Nm	35.6
Polar Moment $J = (\pi * d^4) / (32)$	9.81719E-06
Shear Stress (Pa) $\tau = (T * D) / J$	36.26293172
Angle of Twist $= (T * L) / (J * G)$	0.0001
FS Tresca	8.686556355
FS Von Mises	10.0303713

Gabriel Camacho
 5/20/26
 Group 1: Crema Crew
 Spring 2026
Brew Chamber Analysis

The brew chamber (fig 1) is a highly pressurized section of the assembly and was thus primarily analyzed as a thin walled pressure vessel for ability to maintain pressure without failing. Since it interfaces with food products, it will be made from 316 Stainless steel with properties listed in table 1.

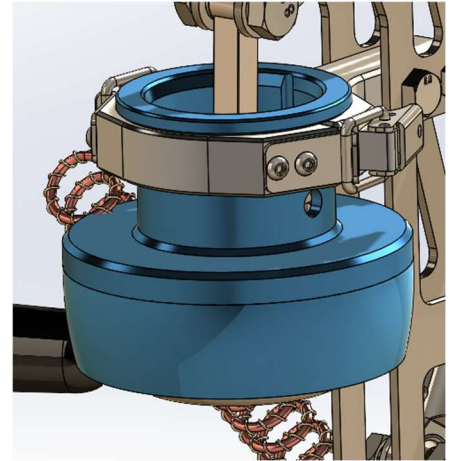


Fig. 1

The highlighted sections show the surfaces of the approximated pressure vessel (fig 2). Hoop and longitudinal stress were calculated assuming a uniform cylinder, since geometry of chamber has multiple irregular parts.

Wall to diameter ratio: $\frac{t}{r_i} = \frac{4mm}{27mm} = 0.148 > 0.2$

Thus, results accurate within 10%

$$\sigma_h = \frac{PD_i}{2t} \quad [5, p. 888]$$

$$\sigma_{h_max} = 1.1 * \frac{900 \text{ kPa}(54mm)}{2(4mm)}$$

$$\sigma_{h_max} = 6.68 \text{ MPa}$$

$$FS = \frac{\sigma_o}{\sigma_{h_max}} = \frac{290 \text{ MPa}}{6.68 \text{ MPa}}$$

$$FS = 43$$

These calculations were conducted for a cylinder that is smaller than the actual chamber, and thus the risk of failure is very minimal.

Stress Analysis		
Material Properties		
AISI 316 Steel		
Tensile Strength	580	MPa
Yield Strength (σ_o)	290	MPa
E (tension)	193	GPa
wall thickness: t	4	mm
inner diameter: r_i	27	mm
thin wall ratio : t/r_i	0.148	
=> accurate within		
10%		
target pressure	900	kPa
hoop stress:	6.075	MPa
10% Range Max	6.6825	MPa
Factor of Safety	43.4	

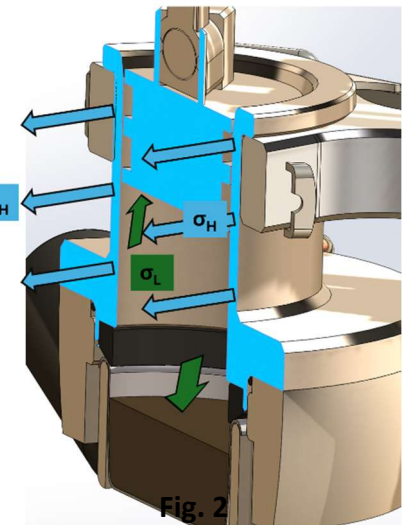


Table 1

Given the set requirements to make an electrical system off 10a of 120V outlet power, the most important mathematical definition will be Ohms law. This is mainly to convert power between different voltages to find our maximum power use cases. This will be observed conservatively looking to maintain 1/3 the max of power supply ratings for longevity by rule a thumb and the peak draw of any device in the system for reliability.

Electrical Analysis – Power



By Joey Dwyer

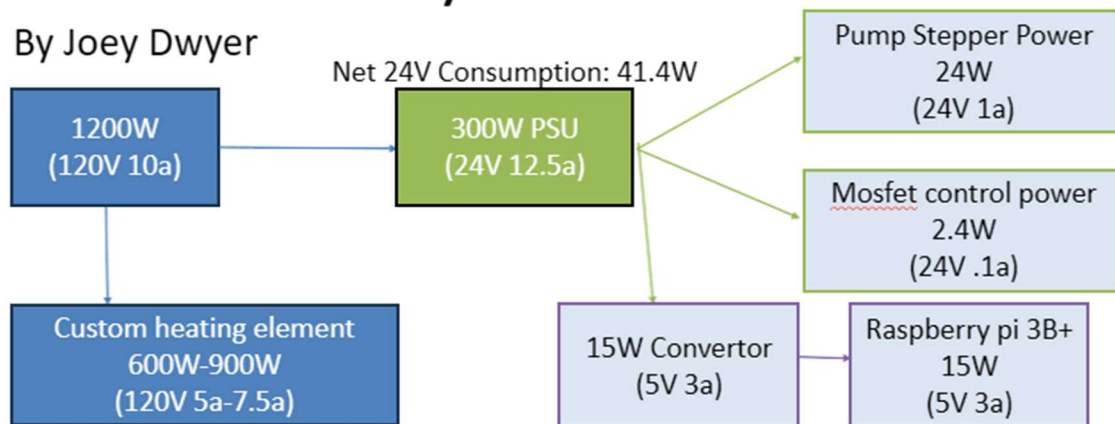


Fig.1 Electrical power breakdown

From the heating analysis we already know the heating element only needs 754w of power which means we have an excess of 146w that can be allotted if needed. With our 300w DC 24V PSU at only 41.4w giving 58w of power before hitting the 100w reliability limit. This allows for future goals to be added down the road like a touch screen system or a motor added to the crank and maybe even a grinding element to the project. However as of now this wiring diagram will outline what features at minimum will be there by the end of 191.

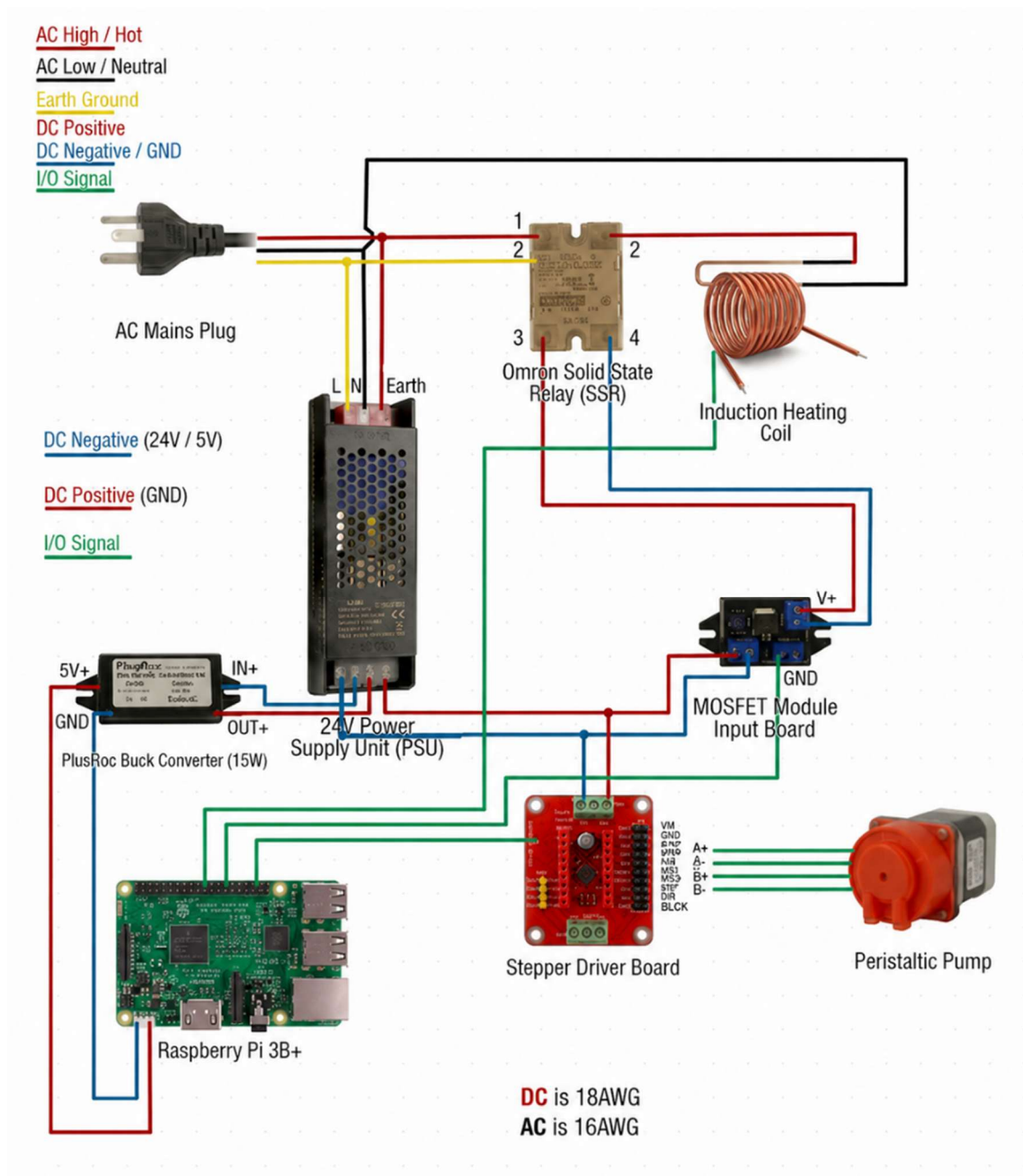


Fig.2 Wiring Diagram

Alexis Phipps
Date: 5/19/26
Group 1: Crema Crew
Spring 2026
O-Ring Analysis

The purpose of the following analysis is to calculate the O-ring crush for the piston seals that interface with the inner cylindrical face of the brewing chamber. This will ensure that the choice of the O-ring including sizing and material choice is reasonable for the scope of the project, and that the seal created is strong enough to not allow any leakage during brewing. The groove channels in the piston head have a dimension of 5mm x 5mm and the piston diameter in the groove area is 43.54mm. The inner diameter of the brewing chamber is 53mm which means that the gap between the piston and the brewing chamber is 0.27mm. The goal is for the O-ring to bridge that gap and fully seal, and if it doesn't, then the piston will need to be sized up. According to Eriks.com [7], the size of the O-ring is based on the inner diameter and the groove channel size. With a groove channel cross sectional area of 5x5mm and an inner diameter of 43.54mm, assuming a dynamic loading scenario with a sectional area approximately 12% smaller than the groove area, the O-ring size is 43 x 5mm with 2.3% stretch. Alternatively, a 42 x 5mm O-ring can be used which has a 3.5% stretch, which is also within the safe range of use as it is less than 5% stretch.

The measurements from the brewing chamber, piston, and groove were input into excel to find the proper size of O-ring. Excel was used to experiment with different standard O-ring sizes and groove depths to narrow down an O-ring that would be suitable to the scenario. The snip from excel is in figure 4 and a sample calculation can be seen below.

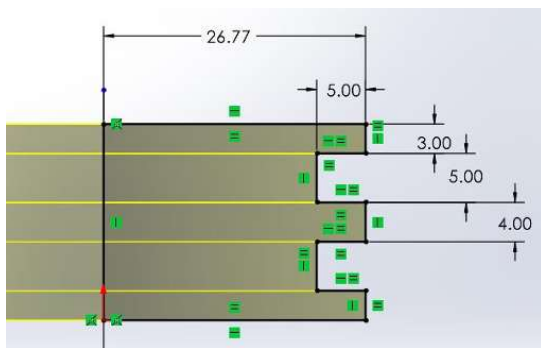


Fig. 3 Measurements of the piston body

Properties (mm)	
Brew Chamber Inner Diam	53
Piston Diameter	52
Groove Depth	4
Groove Diameter, D_{groove}	44
Groove Width	7
O-Ring Diam, W_o	5
O-Ring Inner Diam, D_{ring}	43
Calculations	
% Stretch (1-5%)	2.3
% Squeeze	10

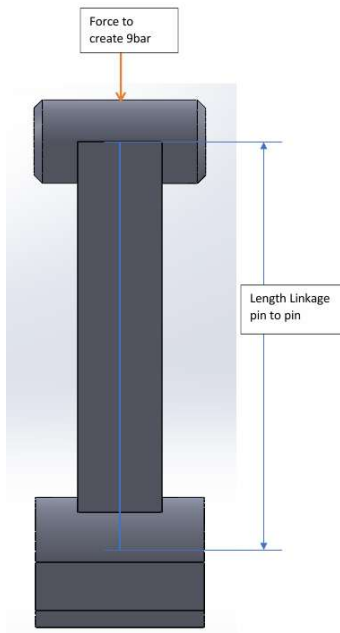
Fig. 4 - Excel Snip showing calculated values using above formula

$$\% \text{ Squeeze} = \frac{W_o - W_c}{W_o} * 100\% = \frac{5 - \left(4 + \frac{53 - 52}{2}\right)}{5} = 10\%$$

$$\% \text{ Stretch} = \frac{D_{groove} - D_{i,ring}}{D_{i,ring}} * 100\% = \frac{44 - 43}{43} * 100\% = 2.3\%$$

Analysis and Buckling Analysis – Piston Linkage

The purpose of the linkage analysis is to understand the forces incurred on the piston linkage during the brewing from the goal 9 bar of pressure existing within the brew chamber during extraction and prove that the linkage will not fail during normal. The body of the linkage is a square 1cm x 1cm that is pinned at each end, made from Aluminum 6061-T6, with a length of 5cm from pin to pin. The mode of failure that would be incurred on this portion is buckling and the calculations and free body diagram are as follows.



$$I = \frac{1}{12}bh^3 = \frac{1}{12} * 0.1 * 0.1^3 = 8.333E - 06$$

$$A = bh = 0.1 * 0.1 = 0.01$$

$$r = \sqrt{\frac{I}{A}} = \sqrt{\frac{8.333E - 06}{0.01}} = 0.02887$$

$$Le = K * L = 1 * 0.05 = 0.05$$

$$\sigma_{cr} = \frac{\pi^2 E}{\left(\frac{Le}{r}\right)^2} = \frac{\pi^2 * 6.89E + 10}{\left(\frac{0.05}{0.02887}\right)} = 2.27E + 11$$

$$\sigma = \frac{P}{A} = \frac{900000}{0.01} = 9E + 7$$

$$SF = \frac{\sigma_{cr}}{\sigma} = \frac{2.27E + 11}{9E + 7} = 2.52E + 3$$

Fig. 5 – Linkage geometry

The calculated factor of safety of this component is 2.52E+3, thus there is no chance this part will fail under normal operating conditions when using this geometry and this material choice.

5. SYSTEM SPECIFICATION AND TEST PLAN

Week	Specification Tested	Target / Acceptance Criteria
6	Static leak integrity of brew chamber	Brew chamber, piston O-rings, fittings, gauge port, and valve connections do not leak before pressurized testing
7	Low-pressure seal performance	Seals hold pressure without major pressure loss
8	Linkage motion under light load	Lever, worm gear, linkage, and piston operate smoothly without binding or permanent deformation
9	Lever actuation force under operating resistance	User can operate the lever with approximately 10 lbf input force
10	Pump flow rate and reservoir capacity	450 mL reservoir supports approximately 10 single-shot cycles including waste fluid
11	Heating coil outlet temperature	Brew water exits heater between 190°F and 205°F
12	Heating repeatability	Heater produces consistent brew temperature over repeated cycles
13	Pressure generation	Brew chamber reaches industry standard espresso pressure
14-16	Full espresso extraction validation	Machine produces 2 oz espresso $\pm 10\%$ in 25 to 30 seconds while repeatedly meeting pressure, temperature, volume, time, and actuation requirements

6. SCHEDULE

Week	Task	Task to be Performed by
0 (Summer)	Order purchased parts including electrical components, pump, pressure gage, grouphead, portafilter, raw stock, bolts/bushings etc.	Gabriel, Joseph, Westley, Alexis
0 (Summer)	Send file for base, uprights, lever arm to SendCutSend for laser cutting	Westley
1	Manufacture brewing chamber and piston head – machined part	Gabriel, Alexis
2	Manufacture piston linkage, piston pin, rod arms – machined parts	Alexis
3	Manufacture hinged clamp components. Test brew chamber fit. – machined parts	Westley
4	Assemble base (bend and weld) with uprights. Insert worm gear and linkage arms and test movements of mechanism. Adjust heim joint as necessary to achieve desired range of motion for piston	All
5	Attach grouphead to brew chamber. Install all O-rings. Test pressure capabilities of brew chamber	All
6	Wrap heating wire around heating tube. Attach tubing to pump and assemble electrical. Test water flow and control of volume.	Josua and Joesph
7	Install wiring and household plug.	Joseph, Josua, Westley
8	Test water heating capabilities with volume control. Attach electrical board	Joseph, Josua, Gabriel

	and pump to base and attach copper tubing to brew chamber	
9	Finish all assembly. Final Testing of full machine workflow and capability.	All

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Espresso Machine - Group 1	
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Vendor Quotes are required for all items listed.

Save all packing slips and receipts for reconciliation

All purchases must be approved

Budget Edit Date 4/29/26

Item #	Category	Part	Vendor	Vendor Part #	Vendor Info/Product Link	Quote #	Unit Cost	Quantity	Cost
1	Structural	Filament 900g Polypropylene spool	Eryone (amazon)	N/A	https://a.co/d/0cQs1CHV		\$32.99	1	\$32.99
2	Structural	Filament 1kg ASA	Overture (amazon)	N/A	https://a.co/d/08ewFaC2		\$19.99	1	\$19.99
3	Structural	Latch	McMasterCarr	6082A16	https://www.mcmaster.com/6082A16/		\$9.05	1	\$9.05
4	Linkage	D-Profile Rotary Shaft	McMasterCarr	8632T38	https://www.mcmaster.com/8632T38/		\$11.83	1	\$11.83
5	Linkage	Oil-Embedded Ball Joint	McMasterCarr	6072K53	https://www.mcmaster.com/6072K53-6072K531/		\$18.32	2	\$36.64
6	Gear	Steel Worm	McMasterCarr	5754SK631	https://www.mcmaster.com/5754SK631/		\$21.79	1	\$21.79
7	Gear	Worm Gear	McMasterCarr	5754SK758	https://www.mcmaster.com/5754SK758/		\$63.68	1	\$63.68
8	Linkage	Flanged Sleeve Bearing	McMasterCarr	6659K117	https://www.mcmaster.com/6659K117/		\$2.09	2	\$4.18
9	Linkage	Flanged Sleeve Bearing	McMasterCarr	6659K224	https://www.mcmaster.com/6659K224/		\$19.43	2	\$38.86
10	Linkage	Rotary Shaft	McMasterCarr	4143N17	https://www.mcmaster.com/4143N17/		\$10.67	1	\$10.67
11	Linkage	Collar	McMasterCarr	6056N16	https://www.mcmaster.com/6056N16/		\$2.25	1	\$2.25
12	Linkage	Flanged Sleeve Bearing	McMasterCarr	6659K146	https://www.mcmaster.com/6659K146/		\$3.26	5	\$16.30
13	Linkage	Collar	McMasterCarr	6058N18	https://www.mcmaster.com/6058N18/		\$6.91	1	\$6.91
14	Structural	10mm bolt (30)	McMasterCarr	91180A634	https://www.mcmaster.com/91180A634/		\$14.90	1	\$14.90
15	Structural	10mm nut	McMasterCarr	90364A102	https://www.mcmaster.com/90364A102/		\$8.00	1	\$8.00
16	Structural	10mm bot (35)	McMasterCarr	91280A371	https://www.mcmaster.com/91280A371/		\$13.20	1	\$13.20
17	Structural	M5 Button Head Hex-Drive Screw	McMasterCarr	92095A207	https://www.mcmaster.com/92095A207/		\$17.80	1	\$17.80
18	Structural	M8 Hex Head Bolt	McMasterCarr	91280A085	https://www.mcmaster.com/91280A085/		\$10.00	1	\$10.00
19	Structural	M8 Hex Nut	McMasterCarr	94168A114	https://www.mcmaster.com/94168A114/		\$6.08	1	\$6.08
20	Structural	M4 Button Head Hex-Drive Screw	McMasterCarr	92095A476	92095a476https://www.mcmaster.com/92095A476/		\$11.55	1	\$11.55
21	Structural	304 SS BasePlate	SendCutSend	N/A	https://sendcutsend.com/		\$200.00	1	\$200.00
22	Structural	304 SS Upright	SendCutSend	N/A	https://sendcutsend.com/		\$25.00	2	\$50.00
23	Structural	304 SS ClampBracket	SendCutSend	N/A	https://sendcutsend.com/		\$40.00	1	\$40.00
24	Structural	Clamp	Westley	WB-1	https://www.linkedin.com/in/westley-baird/		\$0.00	1	\$0.00
25	Piston	Piston Pin	McMasterCarr	89535K17	https://www.mcmaster.com/89535K17-89535K171/		\$4.33	1	\$4.33
26	Piston	Piston Head	McMasterCarr	9535K41	https://www.mcmaster.com/89535K41-89535K412/		\$20.78	1	\$20.78
27	Piston	Piston Linkage	McMasterCarr	9008K12	https://www.mcmaster.com/9008K12-9008K121/		\$6.24	1	\$6.24
28	Piston	Piston O-Ring	Amazon	VUYVUY	https://www.amazon.com/VUYVUY-Ring-Seals%EF%BC%8CSilicone-Width-Gasket/dp/B0F4X4BXQH		\$24.62	1	\$24.62
29	Water	Tank Valve Gasket	In House	NA	Made in house by printing TPU			1	\$0.00
30	Water	Water Gasket	McMasterCarr	9452K114	https://www.mcmaster.com/9452K114/		\$5.43	1	\$5.43
31	Water	Tank lid Pin	In House	N/A	Made in house by printing PETG (part 2)			1	\$0.00
32	Water	Tank lid	In House	N/A	Made in house by printing PETG (part 2)			1	\$0.00
33	Water	Tank	In House	N/A	Made in house by printing PETG (part 2)			1	\$0.00
34	Water	Tank lid Spring	McMasterCarr	9434K58	https://www.mcmaster.com/9434K58/		\$4.70	1	\$4.70
35	Heating	Resistance wire	Amazon	Ni80 wire	Ni80 wire		\$8.80	1	\$8.80
36	Heating	Copper Tubing	Harfington	a19011700ux0032	https://www.harfington.com/products/p-1058691		\$13.58	1	\$13.58
37	Pumping	Paristaltic Pump	Alibaba	NA	Pump		\$17.00	1	\$17.00
38	Brew Chamber	58mm Walnut Portafilter	Flair Espresso	608816113114	Flair 58mm Walnut Portafilter - Flair Espresso		\$72.00	1	\$72
39	Brew Chamber	Etched Puck Screen	Flair Espresso	860012366040	Etched Puck Screen for the Flair 58 Flair Espresso		\$15.00	1	\$15
40	Brew Chamber	4" x 2.5" 316 Stainless Steel Round Bar	Midwest Steel & Alum	316SSRB4	316 Stainless Steel Round Bar Midwest Steel & Aluminum		\$71.99	3	\$215.97
41	Electronics	NTC 3590B 10k ohm thermistor	Uxcell (amazon)	N/A	https://a.co/d/0h9s5Js		\$8.29	1	\$8.29
42	Brew Chamber	Nuova Simonelli Conical head gasket	Espresso Parts	N/A	espressoparts.com		\$3.30	1	\$3.30
43	Brew Chamber	Nuova Simonelli Group Seat	Espresso Parts	N/A	espressoparts.com		\$106.24	1	106.24
44	Brew Chamber	Nuova Simonelli Group Head Bell O-ring	Espresso Parts	N/A	espressoparts.com		\$1.55	1	\$1.55
45	Electronics	120V AC to 24V DC 8.5A 300W PSU	Amazon	LM-HKDY-24V300W-1P	https://a.co/d/04FIDTkG		\$21.99	1	\$21.99
46	Electronics	Raspberry Pi 3B+ 1GB	Amazon	N/A	https://a.co/d/0c4nC6LW		\$53.58	1	\$50.00
47	Electronics	Omron SSR Relay AC/DC	Amazon	G3NA-225B	https://a.co/d/0hxS47YT		\$22.00	1	\$22.00
48	Electronics	Low voltage control Mosfet	Amazon	B0F6K4LX6Y	https://a.co/d/010s5cW9		\$9.88	1	\$9.88
49	Electronics	15A AC Filtered Inlet	Amazon	N/A	https://a.co/d/08MNVcFv		\$8.59	1	\$8.59
50	Electronics	3 slot wago connector	Amazon	221-413	https://a.co/d/0eUvYTnP		\$7.95	1	\$7.95
51	Electronics	Stepper Controller	Amazon	DRV8825	https://www.amazon.com/dp/B0FF44DF9Q?ref=cm_sw_r_cp_ud_dp_ZCGG5WFBNP1WDWYMG502_4		\$9.99	1	\$9.99
52	Electronics	Thermistor ADC module	Amazon	ADS1115	https://a.co/d/07u5bISB		\$11.99	1	\$11.99
53	Electronics	16 AWG wire	Amazon	N/A	https://a.co/d/07uVx2WS		\$18.99	1	\$18.99
54	Brew Chamber	Water Pressure Gauge 49mm	espressocare.com	1245067	Espresso machine Water Gauge Replacement parts for espresso Machine – Espresso Care		\$32.06	1	32.06
55	Electronics	18 AWG wire	Amazon	N/A	https://a.co/d/05OOJB8B		\$7.88	1	\$7.88
56	Electronics	Insulated fork U connectors	Amazon	N/A	https://a.co/d/0cvQKpmf		\$9.99	1	\$9.99
Total									\$1,335.88